



United International University (UIU)
 Dept. of Computer Science & Engineering (CSE)
MID Exam, Trimester: Fall 2025
Course Code: EEE 2123; Course Title: Electronics
Total Marks: 30; Duration: 1 hour 30 minutes

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

1. A crystal of Silicon is doped with Boron and arsenic on two different sides (a–b), and an I-V curve is determined for various applied voltages at 20°C, as shown in the graph. Fill in the following table based on the given information. ($n=1.x$; x = last digit of your id. For example if last digit is 7, then $n=1.7$); [6]

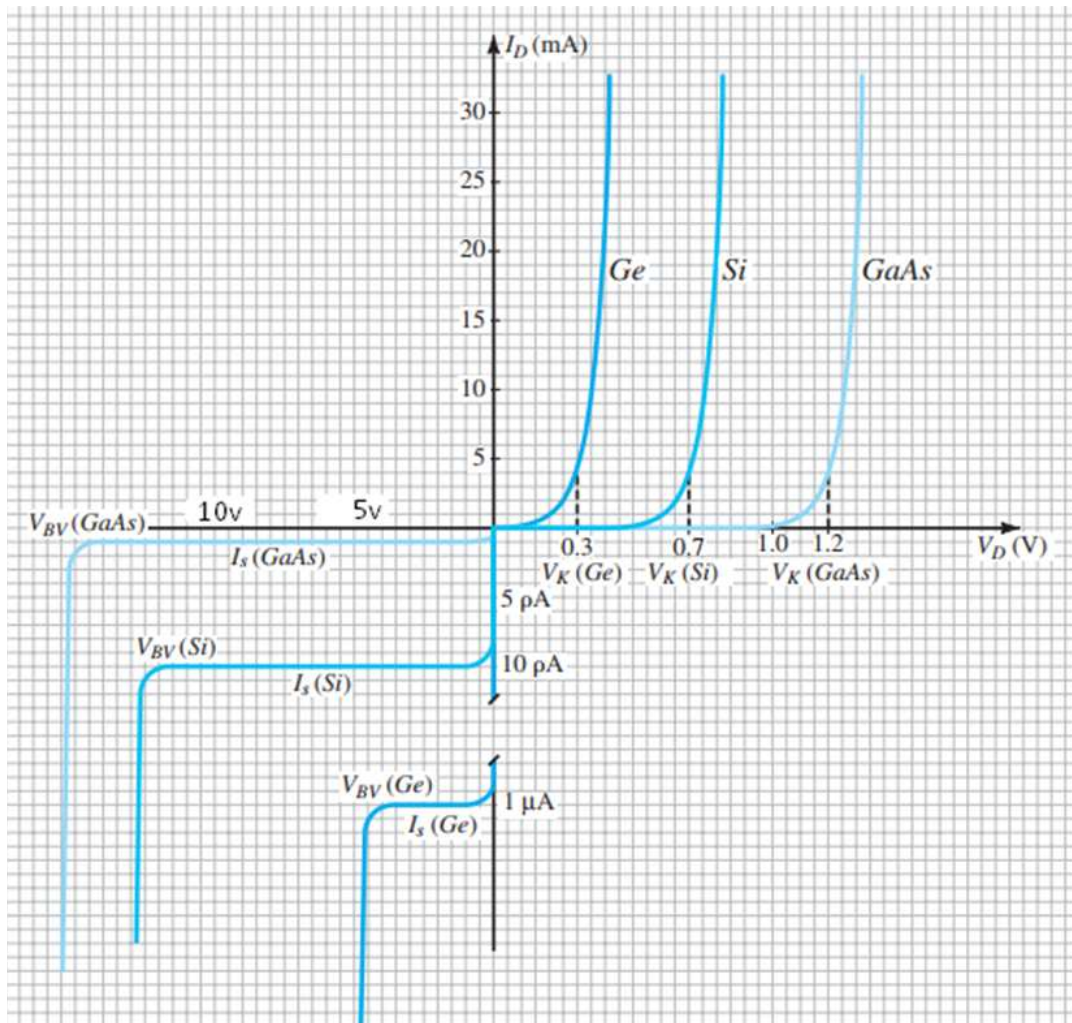


Fig 1: Figure for question 1.

V_{ab}	Biasing Type (Forward/Reverse)	On/Off/Breakdown	Current (a to b)
+3.5v			
+0.1v			
-3.5v			
-15v			

2. Determine $V_1, V_2, V_3, V_4, V_5, I_1, I_2, I_3$ from the following circuit.

[6]

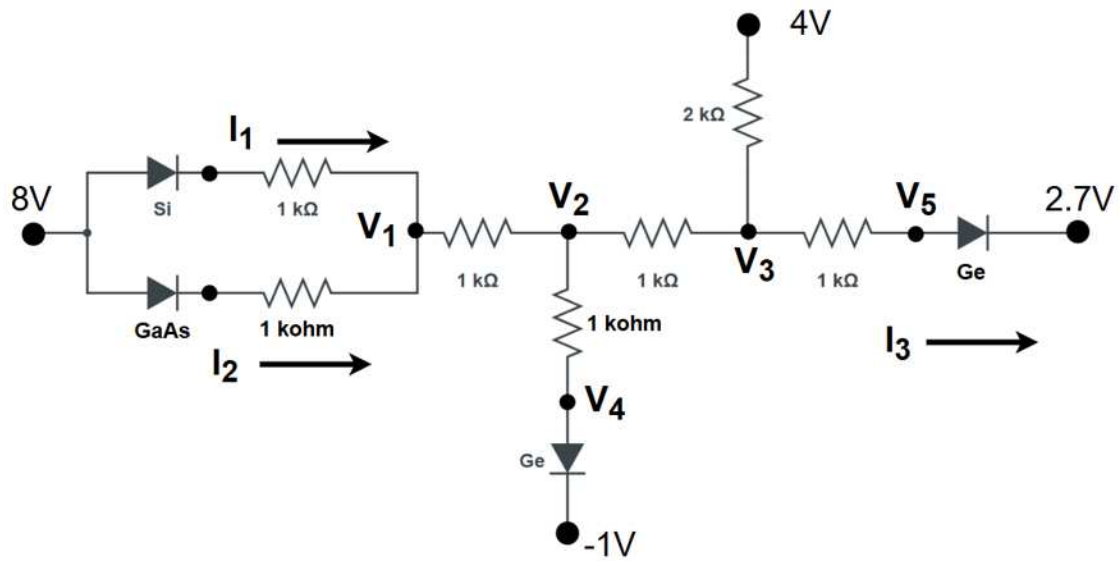


Fig 2: Figure for question 2.

3. The input is a sinusoid with 10V amplitude. Answer the following:

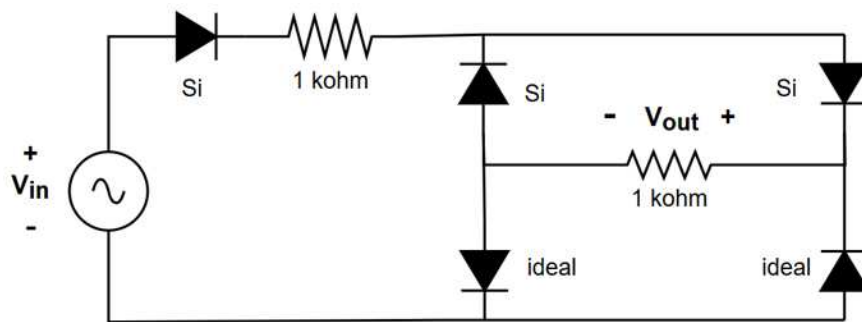


Fig 3: Figure for question 3

- Determine the exact relationship between input and output. [2]
- Draw the waveform of output voltage with proper labeling and values. Is the circuit a half-wave or a full-wave rectifier? Also find the average value of the output. [2]
- Determine PIV of any one of the ideal diodes [2]

4. A 10 ohm resistor has a safe operating range of (-1.45 V to +1.45 V). The following is a good design for such a circuit. A sinusoid of amplitude 10 V is applied at the input terminal.

- Sketch V_o vs Time with arbitrary period for the input voltage. [3]
- Determine the exact values of V_i when Diodes turns on or off. [3]

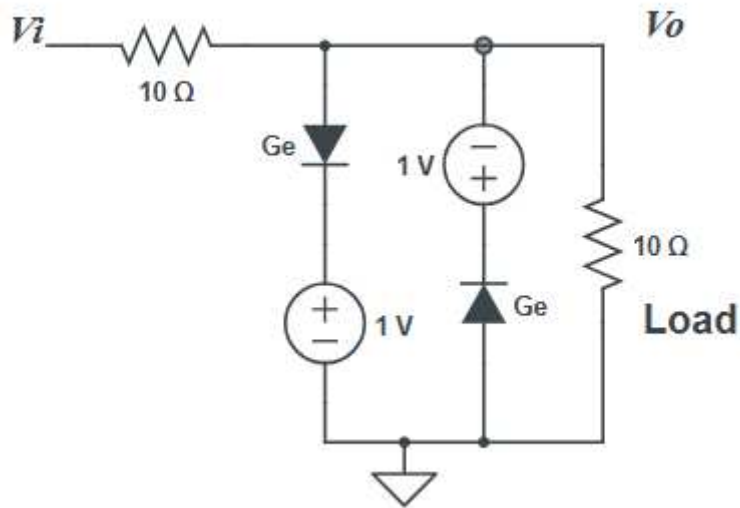


Fig 4:Figure for question 4

5. Look carefully at the following diagram. The input is passed through Circuit 1 and Output 1 is obtained. Similarly, Output 2 is got by passing Output 1 through Circuit 2. Now your task will be to design the both circuits with proper value of each component. The diodes are made of Ge. [3+3]

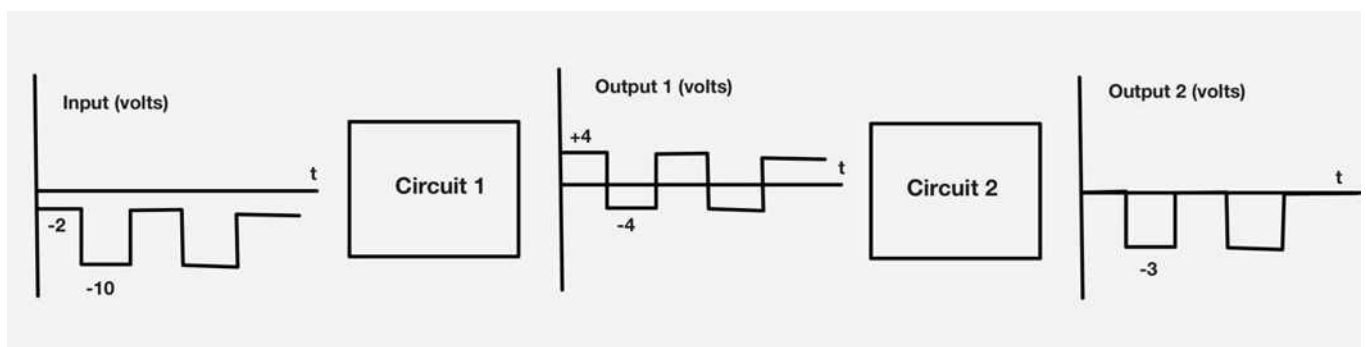


Fig 5:Figure for question 5